

Luca Würker

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Personal Profile

An ambitious and research-oriented mathematician with extensive experience in both academic and industrial mathematical modelling. Currently enrolled in the MSc in Mathematics and Finance at Imperial College London, utilizing advanced numerical methods and stochastic calculus to solve complex problems. Despite success in financial applications, core intellectual drive remains rooted in the rigorous analysis of deterministic systems. Building on successful research in robust statistical estimation and neural ODEs, the primary aim is to transition from financial mathematics back to the fundamental study of applied and computational analysis through the Cambridge MAST Part III.

Education

Imperial College London 09/2025 – 05/2027

MSc Mathematics and Finance (Taught by Department of Mathematics)

- Relevant modules: Stochastic Processes, Statistics, Simulation methods, Numerical methods, Quantum Computing, Stochastic Control, Reinforcement Learning among others
- Awarded a €6000 scholarship by Quantmade in recognition of research contributions, supporting current postgraduate studies.

RWTH Aachen University 10/2022 – 08/2025

BSc Mathematics with Minor in Computer Science

- GPA: 1.5 (Scale: 1.0 best – 5.0 fail)
- Relevant modules: Numerical Analysis Research seminar, Bachelor Thesis in Numerical Analysis, Numerical Analysis I-III, Analysis I-III, Topology, Functional Analysis, Linear Algebra I-II, Stochastics I-II among others

RWTH Aachen University 2021 – 2022

BSc Electrical Engineering and Business Administration

- Program discontinued, switched to Mathematics

Ernst Kalkuhl Gymnasium 09/2013 – 06/2021

Abitur (A-levels)

- GPA: 1.2
- Awarded the **Abitur Prize** by the German Physical Society (DPG) for outstanding achievements in Physics
- Awarded the **Book Prize** by the DPG for excellence in Physics and Natural Sciences
- Term Abroad in England at Stamford Endowed Schools, A levels: Further Maths, Maths, Business, Psychology

Research experience and projects

Bachelor Thesis – Model Reduction of ODEs using Neural Networks – Grade: 1.0 05/2025 - 07/2025

- Developed and analysed a model-reduction framework for autonomous ODEs with applications to multi-scale systems in cloud microphysics (e.g., droplet nucleation and growth), focusing on approximating the dynamical flow map rather than the governing equations themselves.
- Synthesised ideas from numerical analysis, dynamical systems, and current neural/operator learning literature to construct flow-approximating architectures that preserve structural properties essential in physical modelling, such as the semigroup relation of autonomous dynamics.
- Performed a comparative stability and error analysis for stiff and non-stiff regimes relevant to atmospheric physics, testing the reduced models against high-order implicit and explicit Runge–Kutta schemes and analysing their behaviour in fast–slow, highly damped components typical of cloud processes.

Research Seminar in Numerical Analysis – Grade: 1.0 10/2024 - 02/2025

- Studied Gaussian processes as infinite-dimensional Bayesian priors on function spaces, analysing the role of reproducing-kernel Banach spaces in conditioning on linear differential operators.
- Reviewed recent literature on physics-informed GP regression for PDEs, and implemented this framework to obtain probabilistic solutions of heat equations modelling temperature distributions in semiconductor chips.
- Compared GP-based solvers with classical finite-difference methods, highlighting advantages in uncertainty quantification for noisy boundary data and analysing computational trade-offs in matrix-based implementations.

Quantmade Internship and Working Student 03/2025 - 10/2025

- Analysed scientific research papers on robust statistical estimation—especially M-quantile regression—derived implementable formulations for systematic trading, and presented the resulting approach at the QuantMinds International conference (London, 2025).
- Designed and implemented the core modelling pipeline for an M-quantile-based trading strategy, including model specification, numerical methods, and backtesting; currently traded with capital allocation exceeding €3M.

- Performed empirical investigations of factor correlations, signal robustness, and portfolio-optimisation behaviour using statistical modelling and numerical linear-algebra techniques.

Additional Experience

StudyPDF Co-Founder

08/2024 - 05/2025

- Co-founded a startup leveraging AI to create learning materials (e.g., mind maps, mock exams).
- Gained 15,000+ users with positive feedback in less than six months.

Foundational Mathematics Tutor at RWTH Aachen

09/2023 – 09/2024

- Tutored first-year students in Mathematics and Physics, with a focus on Linear Algebra and Real Analysis, emphasizing rigorous problem-solving and conceptual understanding.

Working Student (IT Consulting) at KPS AG

10/2022 – 07/2023

- Supported SAP implementation and data analytics to improve process efficiency.

Mathematical Modelling and Computational Projects

Mathematical Lab – Grade: 1.0

04/2024 - 09/2024

- Translated interdisciplinary real-world problems (e.g., image processing, numerical integration) into formal mathematical frameworks for computational analysis.
- Implemented solutions in C++, developing expertise in applied analysis and computational efficiency.

Practical Studies for Mathematicians in Informatics – ungraded

04/2024 - 09/2024

- Designed and implemented heuristics for the Cops and Robbers pursuit-evasion game on graphs, exploring combinatorial strategies and graph-theoretic concepts.
- Applied algorithmic optimisation to analyze game dynamics and strategic interactions on complex networks.

Skills, Interests, and Volunteering

Languages: German (native), English (fluent), Norwegian (fluent), Spanish (A2)

Technologies: Python, C++, LaTeX, Java, Julia, R, Haskell, Prolog, TypeScript

Interests: Chess, Swimming, (Beach-) Volleyball, Philosophy

Volunteering: Beach Volleyball Trainer for university students (08/2024 – 08/2025) – coached and mentored students, developing teamwork and leadership skills